Parameter-scheduled active chatter control of thin-walled milling: a mixed-μ full-domain gate and an adiabatic whole-pass stability certificate

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Draft of August 24, 2026

Abstract

Thin-walled milling couples a lightly damped, position-dependent structure to a regenerative delay, and active-control studies commonly compress the resulting spatio-temporal variation of the dynamics into a static uncertainty set. Building on the plate/actuator benchmark of Du et al. [Int. J. Mech. Sci. 274 (2024) 109257], we develop a parameter-scheduled active control framework that exploits the variation directly and certifies stability over the entire operating domain. The paper makes four contributions. (i) *Model:* a parameter-scheduled active control family for the varying plant, with the actuator-protection filter *inside* the design model (augmented LQR in scaled coordinates), shaped by measured structural evidence—including the finding that the unfiltered scheduled-LQG performance of the baseline study is mortgaged to unimplementable 1.4–8.8 MHz compensator poles (a performance gap of 0.62 in the shared objective). (ii) *Certification on the physical operating path instead of a single conservative frozen uncertainty set:* real parameters are priced as real—on a plate with $\zeta \approx 0.3\%$, treating them as complex lets stiffness uncertainty masquerade as negative damping and overprices the gate by 4–8× across the whole scheduling domain. A mixed real/complex μ re-reading (Fan–Tits–Doyle bound with solver-independent numerical verification of the returned scales) meets the gate at every one of 21 scheduling nodes for both the benchmark's own controller (peak $\mu_{\mathrm{RS}} = 0.419$) and a realizable scheduled member (0.383), where the complex reading had reported 1.16–85.8; quantities constant along a pass are sliced, and the quantities that actually evolve are chained along the path. (iii) *The finite-cell chain theorem*, proved together with a delay-cover proposition (one certificate covers every shorter delay, hence all faster spindle speeds) and an elementary frequency-domain equivalence between the delay-independent LK inequality and the H_∞ contract of the rank-one regenerative channel: local Lyapunov–Krasovskii certificates, jump factors and a dwell rule combine into an exponential whole-pass certificate whose output is a certified traversal speed v_{cert}—the fastest admissible evolution of the process state under which the certificate remains valid. (iv) *The differential theorem*, proved: the continuous limit for C^1 Riccati certificate fields, governed by $\sigma(x) = \|P^{-1/2}P'P^{-1/2}\|$; its hypotheses are verified computationally on the benchmark. The numerical campaign then *demonstrates* the theory: the realizable member is certified over the full parameter box with $v_{\mathrm{cert}} = 79.9$–82.7 mm/s against an actual feed of 4.90 mm/s—a certificate *rate margin* of ≈16× on the schedule speed, not a productivity figure—and 17.8 mm/s (3.6×) with the removal schedule moving; the filtered gate-crossing member is certified at full box down to $h = 0.104$ mm cell size at $a_{\mathrm{p}} = 0.15$ mm. En route, a quantitative empirical law (modal-direction rotation vs. linewidth; feasible cell diameter scales with ζ) explains why cell-wise LMIs cannot certify such plants and motivates solver-free point certificates on the stable invariant subspace of a

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bounded-real Hamiltonian, with recorded SDP infeasibility verdicts at single vertices shown to be solver failures; grid refinement is used to falsify an over-optimistic coarse-grid claim, and delay-exploiting (TDC-type) members are classified out of the delay-independent contract by design. A lobe-first redesign round then closes most of the benchmark's frozen-speed depth advantage within the same constraint set (lobe floor $0.340 \to 0.439$ mm against the benchmark's 0.474 mm, the shared robustness bar measured binding on both sides, the member's whole-pass certificate strengthened to 49.5 mm/s), quantifying the residual 7% as the measured price of delay independence on this plant; the family's own delay-exploiting member recovers it and more (floor 0.583 mm, 23% above the benchmark) at the measured price of both the certificate and the robustness gate—each guarantee is priced in depth. Every number is derived from stored, re-checkable certificate matrices.

1 Introduction

Chatter in thin-walled milling arises from the interaction of a flexible, lightly damped workpiece with the regenerative cutting mechanism. Because material is removed as the tool traverses the part, the structural dynamics seen by the cutting zone vary both in space (the position of the tool along the workpiece) and in time (the progressive removal), and they do so *slowly and predictably*: the feed speed and the pass schedule are known. Most robust active-control formulations nevertheless freeze this variation into a single uncertainty set and pay the price of guarding against a worst case that includes physically impossible fast excursions [5, 4].

Du, Liu, Dai and Long [1] provide a carefully instrumented benchmark: a five-mode plate model with piezoelectric actuation and displacement sensing, a robust combined time-delay controller (μ-TDC) designed by D–K iteration, and experimental validation. Their robustness analysis, however, follows the common practice of treating the real parametric perturbations—removed-mass fraction, cutting-force coefficient, damping—as complex when the structured-singular-value gate is evaluated.

This paper asks two questions on that benchmark. First, what does the robustness gate look like when the physics is priced as it is—real parameters read as real? Second, can the known slowness of the spatio-temporal variation be converted into a stability *guarantee* over an entire machining pass, rather than a collection of frozen-point statements?

Contributions. The paper is built around four contributions; the numerical campaigns then serve as the *demonstration* of the theory on a validated plant, not as the contribution itself.

1. *A parameter-scheduled active control model* for the milling system with in-process varying dynamics (Section 4): a controller family scheduled by tool position whose actuator-protection filter lives *inside* the design model (augmented LQR solved in scaled coordinates), shaped by measured structural evidence—a performance ceiling $J \approx -0.28$ for filtered structures with its named challenger (phase-lead compensation) tested and rejected, the certainty-equivalence failure of the series-filter variant, and the finding that the unfiltered baseline performance rests on parasitic MHz compensator poles.

2. *Certification on the physical operating path rather than a single conservative frozen uncertainty set* (Sections 3 and 5): real parameters are priced as real (mixed real/complex μ with a verification protocol that removes the SDP solver from the trust chain; measured on the benchmark: the complex reading overprices the gate by 4–8$\times$ across all 21 scheduling nodes, Fig. 2), quantities that are constant along a pass are sliced, and the quantities that actually evolve are chained along the path.

3. *The finite-cell chain theorem* (Theorem 1, with Propositions 1 and 2 and complete proofs in Appendix A): local Lyapunov–Krasovskii certificates, jump factors and a dwell rule combine into an exponential *whole-pass* stability certificate whose output is the certified traversal speed v_{cert} of (2)—a physically interpretable quantity: the fastest admissible evolution of the process state under which the certificate remains valid.

4. *The differential theorem* (Theorem 2): the continuous-limit guarantee that ties the variation of the Riccati certificate field to $\sigma(x) = \|P^{-1/2}(x)P'(x)P^{-1/2}(x)\|$, with regimes and boundary jumps; on the benchmark its hypotheses are verified computationally (a single frozen construction regime, pointwise margins at 100% of nodes, σ converged under grid halving; Fig. 3, 4).

Supporting these, the campaign documents an empirical obstruction law (modal-direction rotation vs. linewidth, which motivates the thin-cell limit), solver-free point certificates on the bounded-real Hamiltonian with a documented SDP false-negative, and two methodological practices we consider part of the result: grid refinement used to falsify an over-optimistic coarse-grid claim, and the classification of delay-exploiting TDC members out of the delay-independent contract by design.

Related work and positioning. Three literatures border this paper. *(a) Switched and LPV time-delay systems.* Dwell-time stability of switched delay systems via multiple or clock-dependent LK functionals is well developed [11, 12, 13, 14], and time-varying delays have themselves been treated as switching signals [15]; parameter-dependent LK functionals for LPV delay systems are surveyed in [16]. Relative to this stream, the present contribution is not a new LMI relaxation but a change of *deliverable and regime*: the switching signal is the machine's own geometry (h_k is cell length, the dwell time is h_k/v at feed speed v), so the certificate returns a physical quantity—the certified traversal speed (2)—and, because the application sits in a regime where cell-wise LMIs are structurally infeasible (the ζ-scaling law of Section 5), the chain is driven into a thin-cell/ differential limit with solver-free point certificates, for which we supply the complete stability proofs (Theorems 1–2). Classical gain-scheduling analysis bounds the admissible parameter *rate* [17, 8]; here the rate bound is constructed, certificate-by-certificate, from stored matrices, and the schedule rate is a design datum (the feed), not an abstraction. *(b) Machining dynamics and active chatter control.* Stability lobes and active suppression are mature [4, 5]; for thin-walled parts specifically, the in-process variation of the dynamics is well recognised and has been handled by worst-position modal design [19], time–space varying PD gains [20], robust LTI envelopes with perturbation models [21, 22], model-predictive formulations [23], and active fixtures with delayed state feedback [24]; the benchmark itself combines a μ-synthesis loop with time-delay control [1]. These works treat the in-process variation through control design (worst-point, robust, adaptive or predictive); what they do not provide—and what the present paper adds—is a stability-and-robustness *certificate along the varying physical operating path*, with a certifiable traversal speed as its output, together with a mixed real/complex reading of the robustness gate. *(c) Computational robustness practice.* The mixed-μ upper bound is classical [2, 3, 10]; the methodological point here is the verification-gated pipeline (no number rests on a solver's verdict) and the documented consequences of omitting it: a modern first-order SDP solver returning "infeasible" on a certificate that exists with a $25\times$ margin.

All computations follow an artifact-first policy: every number quoted here is recomputed from certificate matrices stored on disk, and the complete run logs (including failed rounds and corrected diagnoses) are part of the repository record.

2 Plant, scheduling domain, and fairness protocol

We adopt the benchmark model of [1] without modification: a cantilever-edge thin plate discretised by a Chebyshev–Ritz expansion, calibrated to the measured modal table

$$f = (540,\ 1068,\ 2787,\ 3351,\ 4122)\ \text{Hz}, \qquad \zeta = (0.31,\ 0.17,\ 0.27,\ 0.56,\ 0.35) \times 10^{-2};$$

a piezoelectric patch actuator and a displacement sensor at the dual (collocated-like) corner configuration; and regenerative milling forces with tooth-period delay $\tau = 60/(3\,\Omega)$ at spindle speed Ω (reference condition: $\Omega = 4900\,\text{rpm}$, $a_\text{p} = 0.3\,\text{mm}$, feed speed $4.90\,\text{mm/s}$). Controller synthesis uses the study's two-mode design model; *every* evaluation uses the five-mode truth model.

The scheduling domain is the position $x \in [0, 100]\,\text{mm}$ of the cutting zone along the feed path, discretised where needed at 21 nodes. The physical uncertainty set follows the benchmark: removed-mass fraction $\eta \in [0, 0.0755]$ (calibrated on the +17% frequency drift reported in [1]), cutting-force coefficient $a_4 \in [0.3, 2.9]\,\bar{a}_4$, damping scale $\pm 20\%$, and a mid-pass interpolation state $\xi \in [0, 1]$.

Fairness protocol. All controllers are compared under one protocol: the same plate, patch, sensor and sign convention; the same operating point; the same evaluation (Floquet monodromy with $m = 120$ subintervals on the five-mode model); the same constraints (modulus margin $\max |S| \le 2$, actuation effort $\le 450\,\text{V/N}$); and the same optimiser (particle swarm, 20 particles $\times$ 21 iterations $\times$ seeds $\{1, 2, 3\}$) with a shared objective J (log-margin of the Floquet spectral radius at depth probes 0.3/0.6/1.0 mm). Only the controller structure differs, and its parameter count is reported.

3 Two readings of one robustness gate

The gate quantity is the peak robust-stability structured singular value μ_RS at a scheduling node, actuator block included: the uncertainty structure collects one full complex block for actuator dynamics and repeated *real* scalars for the physical parameters. The benchmark's D–K machinery (and its published numbers) evaluates this with complex readings of the real scalars.

3.1 Why the complex reading overprices this plant

A complex perturbation of a stiffness coefficient contains, at each frequency, a component in quadrature with displacement—i.e. *negative damping.* On a plate whose modal damping is $\zeta \approx 0.3\%$, the open-loop complex-μ of the plate alone is ≈ 40; any controller that must dominate a disguised negative damping across the modes pays an enormous, physically spurious premium. The real parameters of this problem—mass fraction, force coefficient, damping scale—do not have that freedom.

3.2 Mixed real/complex upper bound with verified scales

We evaluate the standard mixed-μ upper bound of Fan, Tits and Doyle [2, 3]: full D-scales on the complex block and Hermitian G-scales on the real blocks, with a bisection on the bound level. Two implementation details matter for trust:

- the non-square actuator channel is squared up by a zero column so the standard LMI form applies;

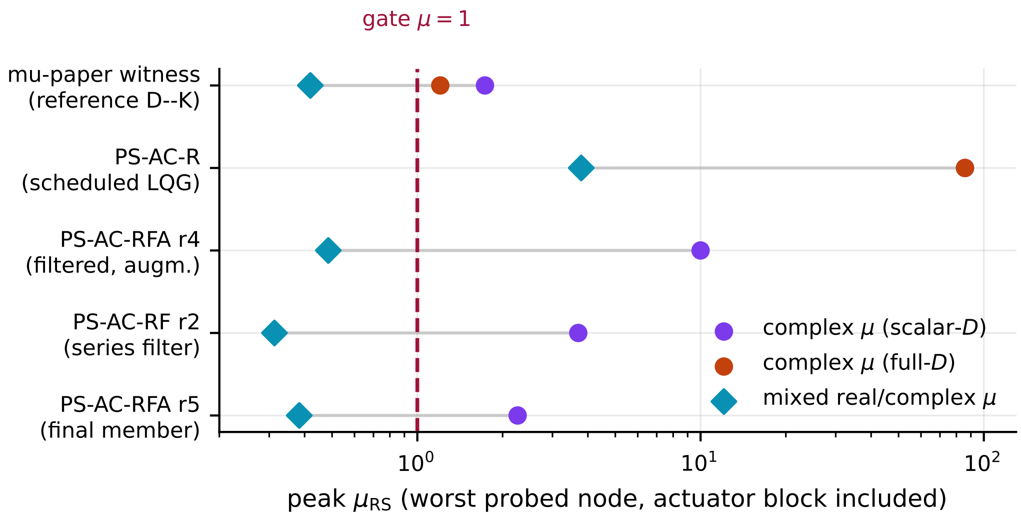

Figure 1: One gate, two prices. Peak μ_{RS} (worst probed node, actuator block included) under the complex readings used by the benchmark's machinery and under the mixed real/complex reading. Every mixed value is a numerically verified upper bound. The complex reading overprices the parametrics by 4–8× on gate-relevant controllers and by 20× on the scheduled-LQG barrier.

- *solver-independent verification*: the returned scales are checked numerically (eigenvalue tests of the bound LMI); an inaccurate solve can only produce a certificate that verifies, or no certificate. The reported value is always a verified upper bound.

At $G = 0$ the bound reduces exactly to the complex bound, so the mixed reading can only tighten. Unit tests hit the known cases (a real block behind an imaginary loop: $2.0 \to\approx 0$; behind a real loop: $2.0 \to 2.0$ exactly; complex blocks: no spurious tightening).

3.3 The gate over the full scheduling domain

Figure 1 places the two readings on one scale for the main controllers of this study; Figure 2 shows the complete 21-node sweep. Three facts organise all downstream design decisions:

1. *The witness passes everywhere.* The benchmark's own stored controller, judged by its own complex machinery at 1.156–1.205 (full-D) and 1.63–1.73 (scalar-D), reads 0.242–0.419 mixed across all 21 nodes—gate met at every node, worst at the free end (1042 Hz).

2. *The scheduled-LQG barrier is real.* The unfiltered scheduled-LQG member reads 3.76–3.79 mixed (complex: 85.8): its actuator-band violence is a genuine physical failure, not a reading artifact. Actuator-aware filtering is therefore necessary, not cosmetic.

3. *"Complex-fragile" can be "real-deep".* A series-filter member whose complex reading collapsed (2.9–3.7) is among the deepest gate-crossers in the mixed reading (0.249–0.313): its fragility was entirely in the reading.

For completeness, the complex-reading gate itself was closed reproducibly before being re-read: a jitter-certified weight search (perturbing frequency grids and weight corners by $\pm\varepsilon$) shows the best *reproducible* complex-reading optimum of the D–K family at 1.728, while apparently better runs at 1.400 are lucky draws whose jitter-certified values are 8.59 and 4.08.

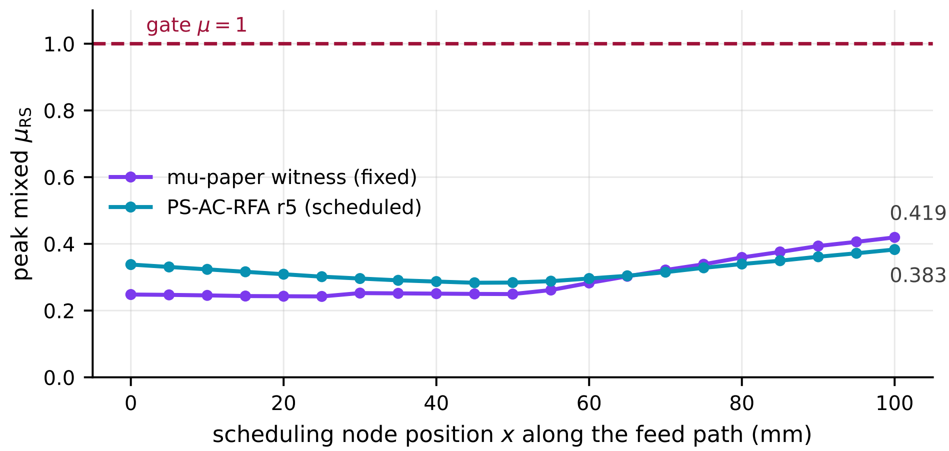

Figure 2: The full-domain gate: peak mixed μ_{RS} at each of the 21 scheduling nodes for the benchmark's fixed controller (witness) and the scheduled member of Section 4. Both curves clear the gate at every node with margin $\geq 2.4\times$; the binding node is the free end, where the removed-mass fraction acts most strongly. The complex reading on the same grid spans 1.63–1.73 (witness) and 2.11–2.27 (member).

3.4 Scheduling the delayed pair is not a lever

The cheapest scheduled attack on the gate–performance question keeps the witness base and schedules only the delayed pair of the TDC law with position, tracking the rank-one regenerative boundary $a_4 D(x)^\top D(x)$ instead of freezing it at midspan (six parameters, same count as μ-TDC). Under the full stage protocol the result is a tie within bisection accuracy, a hair *below* on every axis: certified depth 0.4305 vs 0.4364 mm, margin 1.535 vs 1.543, nominal floor 0.6792 vs 0.6821 mm, mean floor 0.981 vs 1.006 mm. The hypothesis "freezing the pair at midspan loses something material" is falsified by measurement. Combined with item 1 above, this yields the cleanest reading of the benchmark: *under the corrected pricing, the study's own μ-TDC already carries the gate and the performance floors simultaneously*; what scheduling still owes is a floor improvement on a gate-crossing base.

4 An actuator-aware scheduled member, and three structural results

4.1 The filter belongs inside the design model

Because the barrier of item 2 is physical, the scheduled member is synthesised with its actuator-protection filter (three notches at the truncated modes 2787/3351/4122 Hz plus a second-order roll-off) *inside* the design model: an augmented LQR on the joint state (filter; plate) at each scheduling node, with the observer estimating the plate alone, driven by the exact filtered command. Two structural failures motivate this choice, both measured: the series arrangement breaks certainty equivalence (its parametric performance collapsed twice under pressure), and an "observer-consistent" shortcut that rewires the observer around the plain-plant LQR carries an unstable internal observer–filter loop (closed-loop poles at $+2 \times 10^5$ 1/s): separation does not survive lag in the command path.

The augmented Riccati equation is numerically hostile—filter states carry volt-sized signals and plate states metre-sized ones, eight decades apart—and is solved in scaled coordinates (an exact diagonal similarity; the gain is invariant, verified byte-identical where the raw solve succeeds).

The final member (round 5) is an 8-parameter, order-14 controller with fastest pole 5.0×10^4 rad/s (≈ 8 kHz), $M_s = 1.675$, effort 151 V/N; its gate numbers appear in Fig. 2 (point 0.283–0.383 over 21 nodes; worst interpolation cell 0.320), and its nominal position floor at the reference speed is 0.419 mm across the five probe positions ($[0.42, 0.84, 0.72, 1.06, 0.53]$ mm)—above the benchmark reference depth of 0.3 mm.

4.2 Three measured structural results

(a) A J-ceiling of the filtered structure, with its named challenger rejected. Three independent synthesis rounds under different constraint sets converge to $J \approx -0.28$ (seeds: $-0.288/-0.281/-0.326$): the price of actuator-channel robustness for this structure is ≈ 0.42 of J relative to the unfiltered member (+0.145) that fails the gate at 3.8. The one named candidate to break the ceiling—first-order phase-lead compensation inside the design model (10 parameters)—was searched under the identical protocol and rejected: best seed $J = -0.339$, and the optimiser parks both lead parameters at their lower bounds ($f_{\text{lead}} = 424$ Hz at bound 400; $k = 1.56$ at bound 1.5), i.e. it tries to switch the lead off: phase advance is bought with high-frequency gain that the actuator-band constraint immediately bills.

(b) The baseline performance is mortgaged to unimplementable poles. The stored scheduled-LQG member of the baseline study ($J = +0.145$) carries compensator poles at up to 8.8 MHz (family range 1.4–8.8 MHz), originating from both the LQR and Kalman sides. Re-running its own synthesis inside realizability bounds (gain ratio $\leq 10^9$, poles $\leq 2\pi \cdot 20$ kHz) lands at $J = -0.482$ with two agreeing seeds. The gap of 0.63 in the shared objective is the price of implementability for the scheduled-LQG structure and re-prices every performance comparison built on the stored gains. The pathology has a certification-side twin, measured: for an LQG member carrying $\sim$2 MHz parasitic poles the pointwise disc margin is huge (0.05) and a certificate is guaranteed to exist by Proposition 2, yet no floating-point construction we tried can exhibit one—the $\sim 4\times10^3$ closed-loop dynamic range drives the Riccati conditioning past float64 verification floors (`mu_real_cmp.npz`, `adiab_cmp.npz`).

(c) A realizable envelope-law member for certification. For the whole-pass certificate below we also carry `PS-AC-RI`: the same envelope design law inside explicit realizability bounds ($J = -0.482$, $M_s = 1.82$, poles $\leq 1.2 \times 10^5$ rad/s, 5 parameters)—deliberately heavy damping and a tame realization. Judged under the mixed reading it meets the gate as well, at $\mu_{\text{RS}} = 0.688$–0.690 across the probed nodes (complex reading: 15.6), so the certification member of Section 5 carries the gate and the whole-pass certificate simultaneously.

5 The whole-pass certificate

5.1 Chain form: cells, jumps, dwell

The repository's two delay certificates each miss what milling does: the exact crossing test freezes the schedule (each point judged with its own member, nothing certifies the traversal), and a common-P LK certificate prices arbitrarily fast variation the machine never uses. The whole-pass certificate is built instead from the study's own delay-independent LK form given an exponential rate: for cell i with frozen member and vertex family (A, A_d),

$$V_i = x^\top P_i x + \int_{t-\tau}^{t} e^{2\alpha(s-t)} x(s)^\top Q_i\, x(s)\, ds, \qquad \Phi_i = \begin{bmatrix} A^\top P_i + P_i A + 2\alpha P_i + Q_i & P_i A_d \\ A_d^\top P_i & -e^{-2\alpha\tau} Q_i \end{bmatrix} \prec 0. \tag{1}$$

This is the classical delay-independent LK form [6] given an exponential rate; feasibility at (α, τ) covers every delay $\tau' \leq \tau$ (all faster spindle speeds). At a cell switch the state and history are continuous, so $V_{i+1} \leq \mu\, V_i$ for every history iff $P_{i+1} \preceq \mu P_i$ and $Q_{i+1} \preceq \mu Q_i$; the recorded jump factor is the largest generalized eigenvalue over the stored pairs. Monotone traversal with dwell h_k/v then gives the certified traversal speed

$$v_{\mathrm{cert}} = \min_k \; 2\alpha h_k / \ln \mu_k, \tag{2}$$

to be compared with the actual feed speed: the certificate consumes the known slowness of the schedule instead of paying common-P's price. The two statements underlying the chain are the following (proofs in Appendix A).

Proposition 1 (delay cover)**.** *If $P, Q \succ 0$ satisfy $\Phi \prec 0$ in (1) for delay τ and rate $\alpha \geq 0$, then the same pair certifies decay rate α for every* constant *delay $\tau' \in [0, \tau]$ (the tooth period is constant within a pass at fixed spindle speed; shorter delays are faster speeds).*

Theorem 1 (finite-cell chain)**.** *Partition the pass into cells I_k of length h_k and let $(P_k, Q_k) \succ 0$ satisfy (1) with rate $\alpha > 0$ uniformly over the closed convex model hull of cell k (vertex family plus a norm-ε_k residual absorbed by the S-procedure of Lemma 3). For $X, Y \succ 0$ write $\lambda_{\max}(X, Y) = \min\{\mu : X \preceq \mu Y\}$, the largest generalized eigenvalue of the pencil (X, Y); let $\mu_k = \max\{\lambda_{\max}(P_{k+1}, P_k),\, \lambda_{\max}(Q_{k+1}, Q_k)\}$ denote the boundary jump factors, and let the schedule traverse the cells monotonically at speed v. If $\ln \mu_k \leq 2\alpha h_k / v$ at every boundary, then for every parameter path in the tube and every delay $\tau' \leq \tau$ the closed loop is exponentially stable over the whole pass, with*

$$V(t) \;\leq\; \exp\Big(-2\alpha(t - t_0) + \textstyle\sum_{\mathit{switches} \leq t} \ln \mu_k\Big)\, V(t_0);$$

equivalently, every traversal speed $v \leq v_{\mathrm{cert}}$ of (2) is certified.

5.2 Pointwise equivalence, and an empirical obstruction law

Cell-wise SDP solves of (1) refuse on this plant even for near-nominal tubes. A diagnosis ladder isolates why, in two steps.

First, the pointwise tests agree. For rank-one delayed coupling $A_d = b\, c^\top$ (verified: $\sigma_2/\sigma_1 = 1.4 \times 10^{-16}$ for the LQG-type members), the exact crossing measure $\sup_\omega \rho\big((j\omega I - A)^{-1} A_d\big)$ *is* the delay-disc measure $\|G\|_\infty$ of the scalar channel $G(s) = c^\top (sI - A)^{-1} b$: no phase is lost between the crossing test and the DI premise. More is true, by an elementary frequency-domain argument:

Proposition 2 (rank-one KYP equivalence)**.** *Let A be Hurwitz and $A_d = bc^\top$. (a) If $P, Q \succ 0$ satisfy $\Phi \prec 0$ in (1) for some $\alpha \geq 0$, then $\|G\|_\infty < 1$. (b) Conversely, if $\|G\|_\infty < 1$ then, at $\alpha = 0$, the aligned pair $Q = c\, c^\top + \delta I$ (small $\delta > 0$) with P the stabilising solution of a strict bounded-real Riccati equation satisfies $\Phi \prec 0$; by continuity the same holds for all sufficiently small $\alpha > 0$. (Proof in Appendix A.)*

Measured on a refused cell: all 32 vertices pass pointwise ($\|G\|_\infty \in [0.040, 0.541]$), yet the SDP reports infeasible even for a *single* vertex with a 25× margin. Rebuilding that certificate without any optimiser (below) yields a verified functional with margin -5.8×10^{-7}: *the recorded single-vertex SDP refusals are solver failures*, and the verification-gated design of the pipeline ("an inaccurate solve can only produce a certificate that verifies, or none") correctly reported "no certificate" while inviting the wrong mechanism story.

Second, the cell refusals are structural, with a quantitative empirical law. We state this deliberately as a measured observation rather than a theorem: the anisotropy of any certifying storage on this plant (eigenvalue range set by the $\sim 10^4$ frequency ratio of the retained modes) makes a sharp analytic necessity bound unwieldy, while the measurement is unambiguous. The vertex spread inside any practical cell rotates the razor-thin modal eigendirections by many half-power linewidths ($\zeta\omega$): an 8-vertex θ-corner family spreads $\|\Delta A\| = 4 \times 10^{-3}$ against a linewidth of 2×10^{-3} (scaled), and the position axis alone contributes ≈ 700 linewidths per twentieth of the span. A single quadratic storage across eigendirections rotated by more than a linewidth cannot exist; the feasible cell diameter scales with ζ, not with the tube. This replaces our earlier, incorrect naming of the mechanism ("constant scales vs. rank-one coupling"), a correction kept on the record.

5.3 Solver-free point certificates

The cell certificate is therefore rebuilt at a point (and later chained densely). Fix $Q = w\,\overline{A_d^\top A_d} + q_0 s_Q I$ first (aligned with the delayed output directions); by the Schur complement on (1) the coupling term is then a known Riccati channel, and

$$A_c^\top P + PA_c + 2\alpha P + P\Big[\tfrac{s}{e^{-2\alpha\tau}}\,\overline{A_d Q^{-1} A_d^\top} + \kappa\,\Sigma_{\mathrm{in}}\Big]P + (1{+}m)Q + \tfrac{1}{\kappa}\,\Sigma_{\mathrm{out}} + m\,s_Q I = 0 \qquad (3)$$

is solved on the stable invariant subspace of its Hamiltonian (balanced by the exact substitution $P = \theta\tilde{P}$, $\theta^2 = \operatorname{tr} C^\top C / \operatorname{tr} BB^\top$), where the κ-terms absorb vertex spread by a directional Young inequality on the thin-SVD factors of ΔA. *Every* candidate is judged solely by the numerical verifier (eigenvalue tests of Φ, P, Q); nothing rests on the construction. A point certificate costs milliseconds; there is no optimiser in the loop.

5.4 The adiabatic (thin-cell) limit, measured to completion

With point certificates $P(x_k), Q(x_k)$ every h along the pass and scale-balanced neighbour costs $\ln\mu_k = \frac{1}{2}\ln\big(\lambda_{\max}(P_{k+1}, P_k)\,\lambda_{\max}(P_k, P_{k+1})\big)$ (and likewise Q), (2) becomes, as $h \to 0$, $v_{\mathrm{cert}} \to \min_x 2\alpha/\rho(x)$ with rotation density $\rho = d\ln\mu/dx$. Constant-in-time uncertainties are *sliced*—a_4 and ζ do not evolve during a pass, so each realization keeps one chain and the guarantee is the minimum over slices—while the evolving quantities (x, and the removal pair η, ξ) are chained. Four measurement campaigns close the limit:

(i) Slices and rate optimisation. Across all nine $a_4 \times \zeta$ slices, with the decay rate α ladder-optimised per slice, the realizable member at $a_{\mathrm{p}} = 0.10$ mm certifies at $v_{\mathrm{cert}} = 82.7$ mm/s (worst slice, converged under refinement). Against the actual feed of 4.90 mm/s this is a rate margin $v_{\mathrm{cert}}/v_{\mathrm{feed}} \approx 16.9$: the physical schedule evolves 16.9× more slowly than the fastest evolution the certificate tolerates. We stress the interpretation: this is a robustness margin on the schedule rate, *not* a claim that material removal could be increased 16.9-fold. The ζ-slice rows coincide; a wiring check confirms ζ reaches the model ($\|A(1.2) - A(0.8)\| = 2.2 \times 10^{-3}$), so this is genuine certificate insensitivity to ±20% damping.

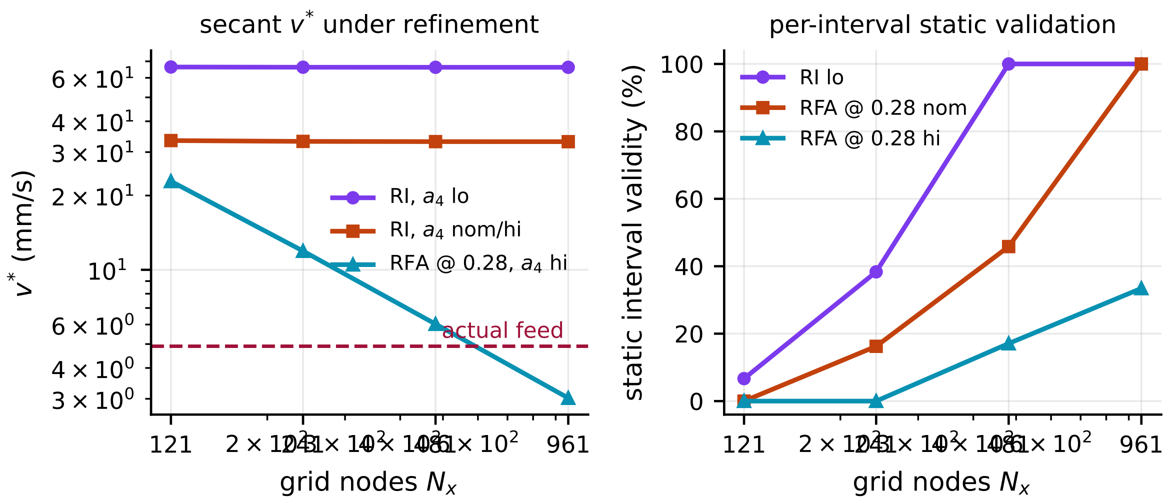

Figure 3: Grid refinement as an honesty instrument. Left: secant v^* under refinement—the realizable member converges on every slice, while the filtered member's a_4-hi slice at $a_\mathrm{p} = 0.28\,\mathrm{mm}$ *diverges* ($22.9 \to 3.0\,\mathrm{mm/s}$, below the actual feed): the coarse-grid certificate was over-optimistic and is corrected. Right: fraction of intervals whose stored functional statically certifies their left/mid/right models; 100% means a fully verified finite chain with no limit argument left open.

(ii) Static validation of finite chains. Upgrading the $O(h^2)$ residual argument to measurement: an interval is *statically valid* if its stored functional certifies the models at its left, mid and right samples. The realizable member's a_4-lo slice reaches 100% validity at $h = 0.104\,\mathrm{mm}$ (Fig. 3, right)—a fully verified finite chain. The nom/hi slices would need $h \sim 10^{-8}\,\mathrm{m}$ (margin/ρ) and stand instead on the differential reading below.

(iii) A correction forced by refinement. The filtered member's full-box certificate at $a_\mathrm{p} = 0.28\,\mathrm{mm}$ does not survive refinement: its a_4-hi slice diverges ($v^* : 22.9 \to 3.0\,\mathrm{mm/s} <$ feed as $N_x : 121 \to 961$), with ρ localising into a spike—the depth sits on the member's exact crossing edge ($a_\mathrm{p}^\infty = 0.2841\,\mathrm{mm}$). The full-box *statically validated* depth is $a_\mathrm{p} = 0.15\,\mathrm{mm}$ (100% validity at $h = 0.104\,\mathrm{mm}$, $v^* = 24.7\,\mathrm{mm/s}$, $5.0\times$ feed). The ρ-spike divergence doubles as a diagnostic for near-marginal corners that coarse grids flatter.

(iv) The live removal tube. Rotation densities along the removal axes are measured directly ($\rho_\eta \approx 645$ for the realizable member—removal motion costs as much as position motion—and $\rho_\xi \approx 1.9$), giving the composite path bound $v^*_\mathrm{path} = \min_x 2\alpha/(\rho_x + \rho_\eta\,\Delta\eta/\ell + \rho_\xi/\ell)$: with the entire admissible removal in a single pass (worst case), $17.8\,\mathrm{mm/s}$—still $3.6\times$ the actual feed.

5.5 The differential theorem, hypotheses measured

Theorem 2 (differential adiabatic chain)**.** *Let the schedule follow the pass at speed v and, on each regime interval, let $x \mapsto (P(x), Q(x))$ be a C^1 field of positive definite matrices satisfying the pointwise LMI (1), for the slice model $(A(x), A_d(x))$, with rate α (in our construction: the frozen-parameter Riccati field of (3), which is real-analytic wherever the Hamiltonian retains its dichotomy [18]). The functional evaluates both matrices at the* current *schedule point, $V(t) = x^\top P(x(t))\,x +$*

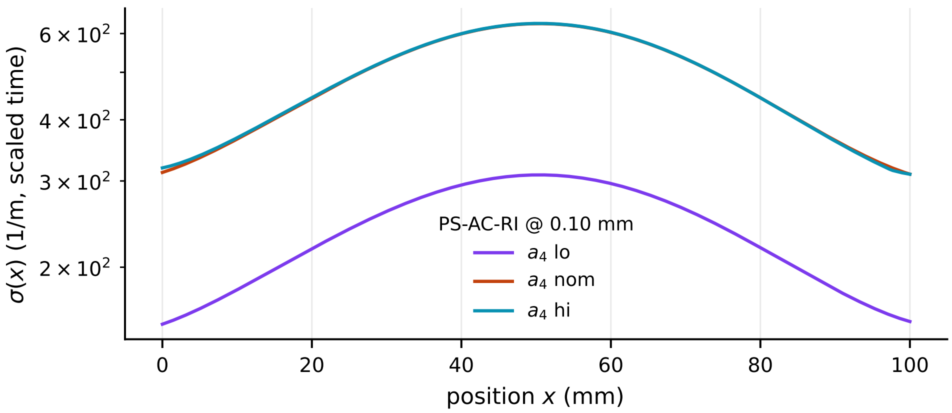

Figure 4: The differential rotation cost $\sigma(x)$ of the frozen-parameter Riccati chains for the realizable member (three a_4 slices, $h/2$ grid). σ is converged under grid halving (ratios 0.96–0.98) and the pointwise LMI margins are re-verified at 100% of the nodes: every hypothesis of Theorem 2 is measured.

$\int_{t-\tau}^{t} e^{2\alpha(s-t)} x(s)^\top Q(x(t))\, x(s)\, ds$. Define $\sigma(x) = \max\left(\|P^{-1/2}P'P^{-1/2}\|_2, \|Q^{-1/2}Q'Q^{-1/2}\|_2\right)$, and let regime boundaries carry jump factors μ_b as in Theorem 1. Then V of (1) satisfies $\dot V \le (-2\alpha + v\,\sigma(x))\,V$ inside regimes and $V^+ \le \mu_b V$ at boundaries; consequently the whole pass is exponentially stable whenever $v < \min_x 2\alpha/\sigma(x)$ and the dwell rule holds at each boundary; by Proposition 1 the statement again covers every constant delay $\tau' \le \tau$. (Proof in Appendix A.)

Measured on this plant (Fig. 4): the winner-parameter census returns a *single* frozen construction per slice across the whole domain (61/61 nodes—no regimes, no jumps); pointwise margins verify at 100.0% of the fine-grid nodes on every slice of both members; and σ is converged under $h \to h/2$ (ratios 0.958–0.984). The theorem-grade speeds, $v^*_{\text{diff}} = 79.9\,\text{mm/s}$ (realizable member, $a_\text{p} = 0.10\,\text{mm}$, worst slice) and 23.6 mm/s (filtered member, $a_\text{p} = 0.15\,\text{mm}$), confirm the secant estimates within $\approx 3\%$.

5.6 Scope: delay-exploiting members

TDC-type members *use* the delay by design (their delayed pair is part of the law, and their delayed coupling is genuinely rank-two: $\sigma_2/\sigma_1 = 0.56$), so "stable for every $\tau' \le \tau$" is not their contract; they rightly fail the pointwise disc at 46–100 of 121 nodes. Their whole-pass statement belongs to the exact-crossing instrument; the certificate developed here covers the delay-agnostic (LQG-type) members.

6 Summary of certified results

Table 1 collects the headline numbers; they are the *demonstration* of Theorems 1–2 on a validated plant, not the contribution itself.

Baseline comparison of certification routes

Table 2 makes the value case directly: the same plant and the same controllers, judged by the four certification routes a practitioner could take. The frozen conservative-set route (the benchmark's

Table 1: Final certified numbers (actual feed 4.90 mm/s; every row recomputable from stored matrices). Ratios $v_{\mathrm{cert}}/v_{\mathrm{feed}}$ are robustness margins on the schedule rate—how much faster the process state could evolve with the certificate still valid—not productivity figures.

Quantity	Value	Rate margin	Basis
Gate, witness, 21/21 nodes	$\mu_{\mathrm{RS}} \le 0.419$	$\ge 2.4\times$	mixed μ, verified scales
Gate + cell, scheduled member	0.283–0.383; cell 0.320	$\ge 2.6\times$	idem
Whole pass, realizable, a_{p}=0.10	$v_{\mathrm{cert}} = 82.7$ (79.9 diff.) mm/s	$16\times$	9 slices, α-opt., converged
with live removal schedule	17.8 mm/s	$3.6\times$	worst single-pass $\Delta\eta$
Whole pass, filtered, full box	$a_{\mathrm{p}} = 0.15$ mm, v^{*}=24.7 (23.6)	$5\times$	static validity 100% at h=0.104 mm
Performance ceiling (filtered)	$J = -0.281$	—	3 rounds; lead probe −0.339 rejected
Realizability price (sched. LQG)	$\Delta J = 0.63$	—	+0.145 (8.8 MHz poles) vs −0.482

practice) refuses the gate outright under its complex reading; frozen-point analysis under the corrected reading meets the gate at every node but says nothing about any $v > 0$ between the nodes; and the classical cell-wise common-storage route is *measured* here on a cell-size ladder (`baseline_cmp.npz`; nominal slice, three samples per cell, no residual—an upper bound on its feasibility): it is infeasible at every practical granularity, becoming feasible only at $h \le 0.1$ mm for the realizable member (1000 cells over the pass; sample spread 6×10^{-3}, i.e. 3–6 linewidths—an independent probe of the empirical ζ-scaling law of Section 5) and $h \le 2.5$ mm for the more heavily damped filtered member. Moreover, at the granularity where per-cell common storage finally exists, it still yields no traversal statement without jump factors and a dwell rule—that is, without Theorem 1: the classical baseline collapses into the present method exactly where it starts to work.

Head-to-head with the benchmark controller

The controller comparison of this paper is deliberately confined to the one comparison that is beyond fairness dispute: the benchmark's own published controller, on the benchmark's own plant, judged by the shared protocol (Table 3). The comparison is respectful by measurement: under the corrected pricing, μ-TDC already carries the gate and the depth floors simultaneously (Section 3), and scheduling its delayed pair buys nothing (the tie of Section 3)—within its own class the benchmark controller is close to optimal. What the scheduled members add over it is orthogonal: a whole-pass certificate its delay-exploiting structure cannot receive under the delay-independent contract (its own whole-pass evidence remains the frozen-point crossing sweep, full-box peak 0.613 at the reference depth), a realizable member that carries gate and pass certificate together, and—for the filtered member—scenario recovery $3.0\times$ faster at $0.45\times$ the control energy and $0.71\times$ the voltage peak. The price is equally measured and stated: lower certified depth floors (0.284 vs 0.436 mm crossing; 0.419 vs 0.682 mm nominal floor). This is a measured frontier between depth headroom and certified traversal with faster, cheaper recovery—not a domination claim. One marginality is stated rather than hidden: the realizable member's worst-position nominal limit (0.299 mm) sits essentially at the reference depth (0.3 mm), so at that condition it operates with the thinnest depth margin of the three (its time response there remains quiet, 4.1 μm peak, Fig. 6). The lobe-first round below narrows the depth side of this frontier within the same protocol.

Table 2: Baseline comparison: what each certification route delivers on the same plant and controllers. "—" = the route produces no statement of that kind.

Certification route	Gate verdict	Whole-pass statement
Frozen conservative set, complex μ (benchmark practice)	refused: 1.156–1.205 (witness, its own judge); 2.1–2.3 (scheduled member)	— (no certificate at any speed)
Frozen-point mixed μ, 21 nodes	met: 0.242–0.419 / 0.283–0.383	— (no statement for any $v > 0$)
Common quadratic LK storage (classical cell-wise LMI)	n/a	infeasible for 1–500 cells (realizable member) and 1–20 cells (filtered); feasible only at $h \leq 0.1$ / 2.5 mm, and even then requires Theorem 1 to stitch cells
Point-certificate chain + differential limit (this work)	met, full 21-node domain	v_{cert} = 79.9–82.7 mm/s (full box; rate margin ≈16×); 17.8 mm/s live tube; 23.6–24.7 mm/s (filtered member, full box, $a_{\mathrm{p}} = 0.15$ mm)

Stability lobes and time-domain response

Figures 5 and 6 present the two views the machining community reads first, computed with the repository's own validated machinery (the Floquet bisection on the five-mode truth model; the Newmark time integration of the scenario protocol). The lobe chart is the *minimum over the tool-path positions* $\{0, L/2, L\}$—the machine-usable curve for a part whose dynamics change along the feed path—and it is also where the shared protocol showed its one measured blind spot: the objective J probes the design speed only, so the J-protocol members were never asked about the rest of the band, and their lobe floors over 3600–7200 rpm (0.340 mm filtered, 0.232 mm realizable) sit well under the benchmark's 0.474 mm. The lobe-first round below closes most of that gap within the same constraints; the figures carry its gate-passing filtered champion PS-AC-RFA-L (0.439–0.729 mm) next to μ-TDC (0.474–0.802 mm), the realizable member (0.232–0.656 mm), the family's delay-exploiting member (0.583–1.349 mm, dashed—its classification is measured below), and the open loop (0.039 mm worst)—all lifted from the open-loop floor by roughly an order of magnitude. In the time domain at the reference condition the open loop diverges within 0.089 s, while the members hold the response at 2–4 μm with peak actuation of 16–37 V—far inside the 450 V authority.

A lobe-first design round

The deficit above is a property of the design pressure, not of the scheduled structure, and it is treated the way every other gap in this study was treated: measure the binding mechanism, change one declared thing at a time, and let pre-declared bars judge the result at full resolution (Floquet $m = 120$, 37 speeds, min over positions). The bars, declared before the first search: L1, beat the incumbent's floor; L2, beat the benchmark's 0.474 mm; L3, hold the shared constraints—and the standing gate G1 disqualifies any winner it rejects.

Round 1 (objective). The objective is replaced by the lobe-floor functional—the same Floquet-margin functional as J, extended over a nine-speed grid spanning the band at probe depths $(0.5, 0.7, 0.9)$ mm—with each swarm warm-started at its stored J-protocol winner; structure, box, screens, optimiser and seeds are untouched. The filtered floor moves $0.340 \to 0.407$ mm, the

Table 3: Head-to-head with the benchmark controller under the shared protocol. Gate = peak mixed μ_{RS} over the probed nodes (complex reading in parentheses). S1 recovery ratios are relative to μ-TDC (= 1). Sources: `stage3_controllers.pkl`, `stage56.pkl`, `stage78.pkl`, `mu_real*.npz`, `adiab_*.npz`.

	μ-TDC (benchmark)	PS-AC-RFA r5	PS-AC-RI
parameters / order	6 / 12	8 / 14	5 / 6
J (shared objective)	-0.025	-0.281	-0.482
M_s / effort (V/N)	2.00 / 152	1.68 / 151	1.82 / 109
gate, mixed μ_{RS}	0.242–0.419 (1.16–1.21)	0.283–0.383; cell 0.320 (2.1–2.3)	0.688–0.690 (15.6)
crossing a_{p}^{∞} / $\delta_{\max}$	0.436 mm / 1.543	0.284 mm / 0.918	—
nominal floor / mean (mm)	0.682 / 1.006	0.419 / 0.712	0.299 / 0.474
recovery: time / energy / peak	1 / 1 / 1	0.33 / 0.45 / 0.71	0.21 / 0.12 / 0.64
whole-pass certificate	out of DI contract (TDC); frozen-point crossing, peak 0.613	$v_{\mathrm{cert}} = 23.6$–24.7 mm/s, full box, $a_{\mathrm{p}} = 0.15$ mm	$v_{\mathrm{cert}} = 79.9$–82.7 mm/s full box; 17.8 live tube

realizable floor 0.232 $\to$ 0.238 mm, and the binding pressures are then *measured*: the filtered winner stands at its self-imposed corner-envelope bar (corner $M_s = 1.935$ of 2.0) with both shared constraints slack, while the realizable winner stands at its search-box walls (gain ratio 8.999 of 9; poles 1.246×10^5 of the 1.257×10^5 rad/s cap) with corner M_s only 1.09.

Round 2 (parity and walls). The benchmark controller itself measures corner $M_s = 2.119$ on the same eight physics corners—the corner bar imposed on our own member was stricter than what the benchmark achieves. Setting the member's corner bar to the benchmark's own measured value (the shared nominal bar $M_s \leq 2.0$ is untouched) lifts the filtered floor to 0.439 mm, and the shared constraint itself now binds exactly ($M_s = 1.999$)—the same bar the benchmark sits at (2.00). A ten-parameter phase-lead arm under identical pressures measures *worse* (0.302 mm) and is recorded as a negative result. Widening the realizable box in gain and ratio only (the certificate's pole cap unchanged) lifts that class's floor to 0.314 mm and even produces a reference-depth path certificate (9.5 mm/s at $a_{\mathrm{p}} = 0.30$ mm)—but the mixed gate rejects the winner ($\mu_{\mathrm{RS}} = 1.21$–1.23 at every probed node): the realizable box is thereby measured to be *load-bearing for the gate*, not an arbitrary restriction, and the paper's realizable member remains PS-AC-RI.

Round 3 (ceiling). A polish at the judge's own resolution ($m = 120$) on exactly the six binding speeds, three independent seeds, moves the filtered floor by less than 0.001 mm. At equal robustness bars on both sides, 0.439 against 0.474 mm (7.4%, with 31 of 37 speeds at or above the benchmark's floor) is therefore the measured price of the delay-independent, whole-pass-certifiable structure on this plant—the depth side of the Table 3 frontier, narrowed from 28% and quantified at its saturation.

Table 4 carries the champion's full battery. It passes the gate (0.332–0.450), improves the shared objective it was not optimised for ($J = -0.166$ vs -0.281), raises the nominal floor to 0.539 mm, keeps the 3.0× faster scenario recovery (at 0.70× the energy), and its whole-pass certificate at $a_{\mathrm{p}} = 0.15$ mm reads $v^* = 49.5$ mm/s on all three a_4 slices at the *fixed* base rate $\alpha = 10\,\mathrm{s}^{-1}$—double the incumbent's α-optimised 23.6–24.7 mm/s, so the lobe-first push strengthened the certificate rather than spending it.

The remaining corner of the frontier is then measured rather than conjectured. The family's own delay-exploiting member PS-TDC-J—the scheduled base with the grafted delayed pair, six parameters as μ-TDC, designed under the *unchanged* shared protocol—already carries a lobe

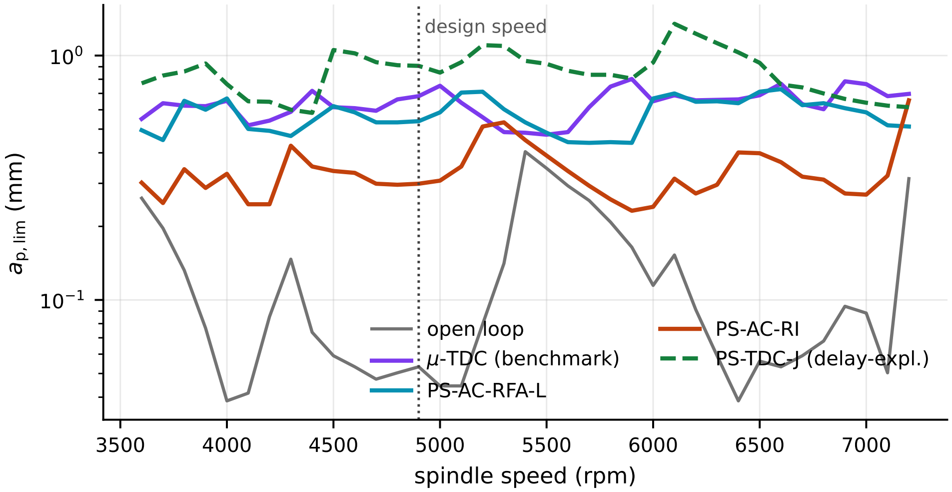

Figure 5: Stability lobes: critical depth of cut vs. spindle speed, minimum over the tool-path positions $\{0, L/2, L\}$ (five-mode truth model, Floquet $m = 120$, bisection to $5\,\mu$m). PS-AC-RFA-L is the lobe-first champion of Section 6; the dashed curve is the family's delay-exploiting member (out of the DI contract and the gate, as marked in Table 4). The dotted line marks the design speed. Sources: `lobes_l2.npz`, `lobes_tdcj.npz`.

floor of 0.583 mm: above the benchmark's floor at all 37 speeds, above the benchmark's own curve pointwise at 31 of 37, with a nominal floor of 0.907 mm (+33% over μ-TDC) and scenario settling 14× faster ($T_s = 6.3$ ms) at 0.79× the energy (1.48× the voltage peak, still far inside authority). Its classification is equally measured: it is out of the delay-independent contract by design, and its unfiltered high-gain base fails the mixed gate outright ($\mu_{\mathrm{RS}} = 3.46$–3.48 at the probed nodes)—exactly the exposure the gate exists to police. Scheduling the delayed pair on the witness base instead extends the tie of Section 3 to the lobe axis (floor 0.472 vs 0.474 mm, pointwise above at only 16 of 37 speeds). The frontier is thereby priced in full on one plant under one protocol: giving up the whole-pass certificate buys the step from 0.439 to 0.474 mm, and giving up the robustness gate as well buys the step to 0.583 mm—each guarantee has a measured depth cost, and each member of the family occupies exactly one declared corner (Fig. 5, dashed curve; Table 4).

Under the corrected pricing the spatio-temporal variation is no longer an adversary to be over-insured against: it is measured, scheduled, sliced where constant, chained where moving, and the guarantee runs over the entire operating domain with an explicit rate margin against the actual feed.

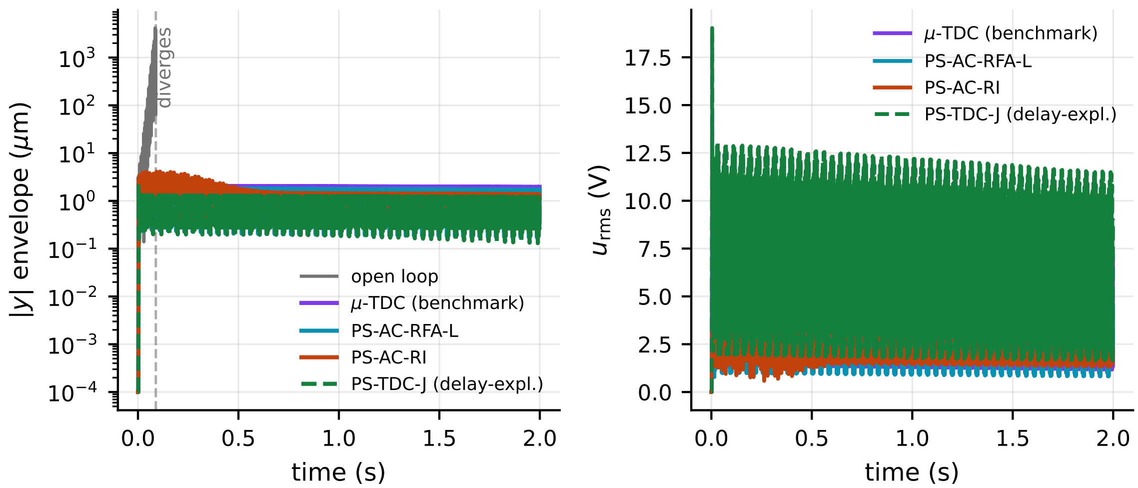

Figure 6: Time response at the reference condition (4900 rpm, $a_\mathrm{p} = 0.3$ mm, moving cut, 2 s window). Left: displacement envelope at the sensor (log scale)—the open loop diverges at $t = 0.089$ s (dashed vertical marker); the members hold a 2–4 μm peak and settle near or below 1 μm. Right: rms actuation voltage of the members. Sources: `timeresp.npz`, `timeresp_l.npz`, `timeresp_tdcj.npz`.

Table 4: The lobe-first round and the priced frontier. Lobe rows are full-resolution verdicts (Floquet $m = 120$, 3600–7200 rpm, min over the tool-path positions); gate and recovery by the standing machinery, each recovery ratio within one measurement batch. PS-TDC-J is the family's delay-exploiting member under the unchanged shared protocol. Sources: `lobes_l2.npz`, `lobes_tdcj.npz`, `mu_real_l.npz`, `mu_real_tdcj.npz`, `s1_l.npz`, `s1_fresh.npz`, `s1_tdcj.npz`, `cert_l.npz`.

	μ-TDC (benchmark)	PS-AC-RFA r5	PS-AC-RFA-L	PS-TDC-J (delay-expl.)
lobe floor, 3600–7200 rpm	0.474 mm	0.340 mm	0.439 mm	0.583 mm
speeds at/above 0.474 mm	37/37	20/37	31/37	37/37
nominal floor / mean (mm)	0.682 / 1.006	0.419 / 0.712	0.539 / 0.886	0.907 / 1.373
gate, mixed μ_{RS} (nodes)	0.242–0.419	0.283–0.383	0.332–0.450	3.46–3.48 (not met)
M_s / effort (V/N)	2.00 / 152	1.68 / 151	2.00 / 160	1.98 / 272
J (shared objective)	−0.025	−0.281	−0.166	+0.156
recovery: time / energy / peak	1 / 1 / 1	0.33 / 0.45 / 0.71	0.33 / 0.70 / 0.98	0.07 / 0.79 / 1.48
whole-pass certificate	out of DI contract	23.6–24.7 mm/s (α-opt), a_p = 0.15 mm	49.5 mm/s (fixed α), $a_\mathrm{p} = 0.15$ mm	out of DI contract (as the benchmark)

7 Limitations and outlook

We state the boundaries of the claims as measured. (1) All results are model-based certifications on the (experimentally validated) benchmark plant of [1]; no new experiments are reported. (2) The nom/hi slices of the realizable member rest on the differential reading of Theorem 2; their static cell validation is physically out of reach ($h \sim 10^{-8}\,\mathrm{m}$). (3) The η/ξ tube enters the headline slice numbers frozen at midspan and enters the path bound through measured densities under a single-pass worst case; a fully coupled (x, η, ξ) chain is mechanical with the same tooling. (4) The decay rate α is grid-optimised, not continuously optimised. (5) The performance objective J is the benchmark's Floquet-margin protocol; other objectives may re-order members but not the certificates. (6) Deep-cut corners near a member's exact crossing edge are intrinsically hostile to the DI premise (the measured ρ-spike); certification there should be attempted at the exact delay only. (7) The lobe-first champion's certificate is reported at the fixed base rate $\alpha = 10\,\mathrm{s}^{-1}$; the incumbents' per-slice α optimisation can only raise it. Its crossing pair $(a_\mathrm{p}^\infty, \delta_\mathrm{max})$ has not been re-measured; the frozen-speed lobe verdicts of Table 4 are the sharper statement of the same axis.

Significance. Beyond the benchmark, the paper argues for three transferable practices. First, robustness gates on lightly damped structures should be reported under the mixed real/complex reading: on this plant the reading, not the physics, decided the gate, and the same inflation mechanism (parametric stiffness masquerading as negative damping) is generic to any $\zeta \ll 1$ structure. Second, when the schedule of an LPV delay plant is slow and known—machining passes, but equally any traversal-type process—the deliverable of a stability analysis can and should be a *certified schedule speed* (Theorems 1–2) rather than a family of frozen-point verdicts; the empirical ζ-scaling law explains when cell-wise LMIs cannot supply it and the point-certificate chain can. Third, certification pipelines should be verification-gated end to end: the documented single-vertex SDP false-negative shows that solver verdicts, positive or negative, are not evidence; stored, re-checkable certificate matrices are.

Reproducibility and data availability

The complete pipeline—plant construction, both μ readings with their verification layer, all synthesis rounds including failed ones, the diagnosis ladder, the solver-free certificates, and every figure of this paper—is scripted in the project repository. All quantities are recomputed from stored artifacts (`mu_real_21.npz`, `adiab_pass.npz`, `adiab_rigor.npz`, `adiab_rigor2.npz`, `adiab_diff.npz`, `baseline_cmp.npz`, `mu_real_cmp.npz`, `adiab_cmp.npz`, `lobes.npz`, `timeresp.npz`, `ri_s1.npz`, `s1_fresh.npz`, and for the lobe-first round `lobes_l.npz`, `lobes_l2.npz`, `mu_real_l.npz`, `s1_l.npz`, `timeresp_l.npz`, `cert_l.npz`, `lobe_champions.pkl`, `lobes_tdcj.npz`, `mu_real_tdcj.npz`, `s1_tdcj.npz`, `timeresp_tdcj.npz`, run logs), and corrections to earlier readings are appended to the record rather than rewritten. The lobe-first members carry the repository keys `ps_ac_rfa_l2` (PS-AC-RFA-L, the champion), `ps_ac_rfa_l`, `ps_ac_rfl_l` and `ps_ac_rfa_l3` (the recorded round-1, lead and polish arms), and `ps_ac_ri_l` / `ps_ac_ri_l2` (the realizable-class arms, the latter gate-disqualified).

A Proofs

Throughout, for a delay $\tau' \in [0, \tau]$ write the functional of (1) as $V = V_P + V_W$ with $V_P = x^\top P x$ and $V_W = \int_{t-\tau'}^{t} e^{2\alpha(s-t)} x(s)^\top Q\, x(s)\, ds$.

A.1 Proof of Proposition 1

Differentiating along trajectories of $\dot{x} = Ax + A_d x(t - \tau')$,

$$\dot{V} + 2\alpha V \;=\; \xi^\top \begin{bmatrix} A^\top P + PA + 2\alpha P + Q & PA_d \\ A_d^\top P & -e^{-2\alpha\tau'} Q \end{bmatrix} \xi, \qquad \xi = \big(x(t),\, x(t-\tau')\big).$$

Since $\tau' \le \tau$ and $Q \succ 0$, $-e^{-2\alpha\tau'} Q \preceq -e^{-2\alpha\tau} Q$, so the matrix above is $\preceq \Phi \prec 0$: hence $\dot{V} \le -2\alpha V$, and $V_P \le V$ gives $|x(t)|^2 \le \lambda_{\min}(P)^{-1} e^{-2\alpha(t-t_0)} V(t_0)$. □

A.2 Proof of Proposition 2

(a) $\Phi \prec 0$ as a real symmetric matrix implies $\xi^* \Phi\, \xi < 0$ for every nonzero complex $\xi \in \mathbb{C}^{2n}$. Fix $\omega \in \mathbb{R}$ and set $\hat{x} = (j\omega I - A)^{-1} b$ (well defined, A Hurwitz), $g = c^\top \hat{x} = G(j\omega)$, $w = \hat{x}^* P b$, and $\xi = \big(\hat{x},\, e^{\alpha\tau} e^{j\varphi}\, \hat{x}\big)$ with φ free. Using $A\hat{x} = j\omega\hat{x} - b$,

$$\hat{x}^* \big(A^\top P + PA\big)\hat{x} = 2\,\mathrm{Re}\big(\hat{x}^* P A \hat{x}\big) = -2\,\mathrm{Re}(w),$$

and $A_d \xi_2 = b\, g\, e^{\alpha\tau} e^{j\varphi}$, while the Q-terms cancel: $\hat{x}^* Q \hat{x} - e^{-2\alpha\tau}\, e^{2\alpha\tau}\, \hat{x}^* Q \hat{x} = 0$. Hence

$$0 > \xi^* \Phi\, \xi = -2\,\mathrm{Re}(w) + 2\alpha\, \hat{x}^* P \hat{x} + 2\, e^{\alpha\tau} \mathrm{Re}\big(w\, g\, e^{j\varphi}\big) \qquad \text{for every } \varphi.$$

If $w = 0$ this reads $0 > 2\alpha \hat{x}^* P \hat{x} \ge 0$, a contradiction unless $\hat{x} = 0$; so $w \ne 0$ wherever $\hat{x} \ne 0$. Maximising over φ, $e^{\alpha\tau} |w|\, |g| < \mathrm{Re}(w) - \alpha \hat{x}^* P \hat{x} \le |w|$, whence $|G(j\omega)| < e^{-\alpha\tau} \le 1$ for every ω; since G is strictly proper and continuous, $\|G\|_\infty < 1$.

(b) Pick $\gamma \in (\|G\|_\infty, 1)$. By the strict bounded-real lemma [9] there exists $P \succ 0$ with

$$A^\top P + PA + \gamma^{-2} P b\, b^\top P + c\, c^\top \;\prec\; 0.$$

Set $Q = c\, c^\top + \delta I$, $\delta > 0$. At $\alpha = 0$ the Schur complement of Φ with respect to its $(2,2)$ block $-Q$ is

$$A^\top P + PA + Q + Pb \left(c^\top Q^{-1} c\right) b^\top P, \qquad c^\top Q^{-1} c = \frac{|c|^2}{\delta + |c|^2} < 1 < \gamma^{-2},$$

which is $\preceq A^\top P + PA + c\, c^\top + \gamma^{-2} P b b^\top P + \delta I \prec 0$ for δ small enough; hence $\Phi \prec 0$. All inequalities are strict, so they persist for all sufficiently small $\alpha > 0$ by continuity of Φ in α. □

A.3 An absorption lemma

Proposition 3 (S-procedure residual absorption)**.** *In the closed loop of* (1)*, suppose the model residual between the true cell continuum and the vertex interpolant is supported on the* plant block*—nonzero only in the plant rows and plant columns of* (A, A_d)*—which holds here because the member, the actuator path and the sensor geometry are frozen within a cell while only the plant physics varies. Write the residual as* $(E_0 \Delta_A S_p,\, E_0 \Delta_{A_d} S_p)$ *with* $\big\|[\Delta_A \;\; \Delta_{A_d}]\big\|_2 \le \varepsilon$*, where* S_p *selects and* E_0 *embeds the plant coordinates. If there exists* $\lambda > 0$ *such that*

$$\begin{bmatrix} \Phi_0 + \lambda F^\top F & \mathcal{E} \\ \mathcal{E}^\top & -\lambda I \end{bmatrix} \prec 0, \qquad F = \mathrm{diag}(S_p, S_p), \quad \mathcal{E} = \begin{bmatrix} \varepsilon\, P E_0 \\ 0 \end{bmatrix},$$

at the interpolant Φ_0*, then* $\Phi(A_0 + E_0 \Delta_A S_p,\, A_{d,0} + E_0 \Delta_{A_d} S_p) \prec 0$ *for every admissible residual.*

Proof. The residual enters (1) affinely:

$$\Phi = \Phi_0 + \mathcal{L}\,\Delta\,F + (\mathcal{L}\,\Delta\,F)^\top, \qquad \mathcal{L} = \begin{bmatrix} PE_0 \\ 0 \end{bmatrix}, \quad \Delta = [\Delta_A \;\; \Delta_{A_d}],$$

since $\Delta F\,\xi = \Delta_A x_p + \Delta_{A_d} x_{d,p}$ reproduces exactly the perturbation's action on the state pair. For any $\lambda > 0$, Young's inequality with $\|\Delta\|_2 \le \varepsilon$ gives

$$\mathcal{L}\Delta F + (\mathcal{L}\Delta F)^\top \;\preceq\; \lambda^{-1}\varepsilon^2\,\mathcal{L}\mathcal{L}^\top + \lambda\,F^\top F \;=\; \lambda^{-1}\mathcal{E}\mathcal{E}^\top + \lambda F^\top F.$$

By the Schur complement, the bordered LMI is exactly $\Phi_0 + \lambda F^\top F + \lambda^{-1}\mathcal{E}\mathcal{E}^\top \prec 0$, which therefore dominates Φ from above. □ □

A.4 Proof of Theorem 1

In-cell decay. The bordered matrix of Proposition 3 is itself affine in (A, A_d) (only Φ_0 depends on the model, affinely), so its negativity at the vertex family implies its negativity at every convex interpolant, and Proposition 3 then absorbs the measured ε_k-residual around each interpolant: the LMI holds at every model the cell continuum can produce. Since (P_k, Q_k) are constant inside the cell, the derivative computation of Proposition 1 applies verbatim with the time-varying model $(A(t), A_d(t))$ evaluated pointwise; hence for every admissible parameter path, $\dot V_k + 2\alpha V_k \le 0$ inside cell k, uniformly in the constant delay $\tau' \le \tau$ by Proposition 1.

Jump. At a switch instant t_k the state and the history segment are continuous. Both terms of the functional are monotone in their matrices: $P_{k+1} \preceq \mu P_k$ gives $V_P^{(k+1)} \le \mu V_P^{(k)}$, and $Q_{k+1} \preceq \mu Q_k$ gives $V_W^{(k+1)} \le \mu V_W^{(k)}$ pointwise under the (positive) exponential weight; with $\mu = \mu_k$ as defined, $V_{k+1}(t_k) \le \mu_k V_k(t_k)$.

Envelope. This is the multiple-Lyapunov-function argument of dwell-time theory [7] carried by LK functionals: between switches $V(t) \le e^{-2\alpha(t-t_k)} V(t_k^+)$; composing across cells,

$$V(t) \le \exp\Big(-2\alpha(t-t_0) + \sum_{\text{switches}\le t} \ln \mu_k \Big) V(t_0).$$

Monotone traversal at speed v spends dwell $h_k/v \ge \ln \mu_k/(2\alpha)$ in cell k, so each factor $e^{-2\alpha h_k/v}\mu_k \le 1$: the envelope is non-increasing across every switch and decays exponentially inside cells; a uniform decay rate follows for v strictly below v_{cert}. Finally $V_P \le V$ converts the envelope into an exponential bound on $|x(t)|$. □

A.5 Proof of Theorem 2

Inside a regime, write $P(t) = P(x(t))$, $Q(t) = Q(x(t))$ with $\dot P = vP'$, $\dot Q = vQ'$ along the traversal. By definition of σ, and since $P, Q \succ 0$,

$$-\sigma P \preceq P' \preceq \sigma P, \qquad -\sigma Q \preceq Q' \preceq \sigma Q.$$

Differentiating $V = x^\top P(t)x + \int_{t-\tau}^{t} e^{2\alpha(s-t)} x(s)^\top Q(t)\, x(s)\, ds$:

$$\dot V + 2\alpha V = \xi^\top \Phi\big(x(t)\big)\,\xi \;+\; v\,x^\top P' x \;+\; v\int_{t-\tau}^{t} e^{2\alpha(s-t)} x(s)^\top Q'\, x(s)\, ds.$$

The first term is ≤ 0 by the pointwise LMI; the second is $\le v\sigma\, x^\top P x$; the third is $\le v\sigma \int e^{2\alpha(s-t)} x^\top Q x$ because the integrand weight is positive. Hence $\dot V \le \big(-2\alpha + v\,\sigma(x(t))\big)V$. Regime boundaries are finitely many and carry the jump argument of Theorem 1; with $v < \min_x 2\alpha/\sigma(x)$ the decay rate $\inf_x(2\alpha - v\sigma(x)) > 0$ is uniform, and the dwell rule handles the boundary factors. □

A.6 Remark: soundness of the numerical verifier

Every certificate in this paper is accepted only after symmetric eigenvalue tests of the assembled Φ, P, Q (largest eigenvalue of $\frac{1}{2}(\Phi + \Phi^{\top})$ below a strict negative tolerance; positivity floors on P, Q). Symmetric eigenvalue computations are backward stable, so the accepted margins dominate the floating-point perturbation at the tolerances used ($|\Phi|$-relative 10^{-8} or better in all accepted certificates); a fully rigorous variant would replace these by interval or rational arithmetic, which the stored matrices make possible *post hoc*.

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